

ALY ASHER

Staff Data Engineer | Principal Data Architect |
Cloud & Healthcare Data Engineering

alyasher92@gmail.com | Valley Stream, New York 11580

SUMMARY

Result driven Staff Data Engineer with over 10 years of experience architecting, developing, and optimizing large scale data platforms across FinTech, HealthTech, E-Commerce, and SaaS domains. Expert in Python, PySpark, SQL, Apache Airflow, Databricks, and AWS (S3, Redshift, Glue, Lambda) with deep expertise in ETL/ELT design, data modeling, and data lakehouse architecture. Proven track record in building real time analytics, machine learning pipelines, and scalable data governance frameworks using Snowflake, Kafka, DBT, and Terraform. Adept at leading cross functional engineering teams, modernizing legacy systems, and driving performance improvements exceeding 40%. Committed to delivering high impact, cloud native data solutions that enable strategic decision making, operational efficiency, and business growth in fast paced enterprise environments.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Scala, Java, R, Shell Scripting, Bash, JavaScript

Data Engineering & Big Data: Apache Spark, PySpark, Hadoop, Hive, HDFS, Kafka, Flink, Kinesis, NiFi, Airflow, DBT, Databricks, AWS Glue, AWS Lambda, Step Functions, EMR, Snowflake, Delta Lake, Iceberg, Hudi, Data Lakehouse Architecture, ETL/ELT Pipelines, Batch & Streaming Data Processing, Data Lineage, Data Quality Frameworks (Great Expectations, Deequ), Metadata Management

Databases: PostgreSQL, MySQL, SQL Server, Oracle, MongoDB, Cassandra, DynamoDB, Redis, Elasticsearch, BigQuery, Redshift, Snowflake, Data Modeling, Query Optimization, Indexing, Partitioning, Caching, Performance Tuning

Cloud & DevOps: AWS (S3, EC2, RDS, Redshift, Glue, IAM, Lambda, Lake Formation, CloudWatch), Azure, Google Cloud Platform, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, GitLab CI/CD, Ansible, CloudFormation, Linux Administration, Observability (Prometheus, Grafana), Logging (ELK Stack), Monitoring & Alerting, Security Automation

Healthcare Data: HL7, FHIR, HIPAA Compliance, PHI Data Handling, EHR/EMR Integration, Clinical Data Warehousing, Claims & Encounter Data, Population Health Analytics, Medical Coding Standards (ICD-10, CPT), Data De-identification & Encryption, Healthcare Data Governance, Clinical KPIs & Outcomes Analytics, Healthcare Data Interoperability

Data Analytics: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Looker, Power BI, Tableau, AWS QuickSight, Data Visualization, Business Intelligence (BI), OLAP Cubes, Star & Snowflake Schema Design, KPI Dashboards, Statistical Analysis, Time Series Analysis, Reporting Automation, Experimentation & A/B Testing

Data AI & Machine Learning: Scikit-learn, TensorFlow, PyTorch, MLflow, SageMaker, Feature Engineering, Model Deployment, Predictive Analytics, Anomaly Detection, MLOps, Generative AI, LLM Integration, Natural Language Processing (NLP), Recommendation Systems, Forecasting Models, Explainable AI (XAI), Data-Driven Decision Systems

Tools & Collaboration: Git, Bitbucket, Jira, Confluence, Slack, Trello, Asana, Agile, Scrum, Kanban, TDD, API Integration, REST, GraphQL, JSON, Parquet, Avro, ORC, Data Version Control (DVC), CI/CD Automation, Security & Compliance (HIPAA, GDPR, SOC 2), Documentation & Data Cataloging (Alation, Collibra, AWS Glue Catalog)

EXPERIENCE

Staff Data Engineer (Cloud-Native & SaaS Platforms)

07 / 2023 – Present

BetterCloud

- Led the end to end design of a multi tenant cloud data platform using Databricks, Snowflake, and AWS, serving analytics for global SaaS products.
- Built a modular ETL/ELT framework using DBT and Airflow, reducing pipeline maintenance overhead and failure rates by 60%.
- Managed petabyte scale data ingestion using Kafka, Flink, and S3, enabling near real time insights for usage metrics and application telemetry.
- Implemented data governance and access policies using AWS Lake Formation and IAM, ensuring compliance across distributed environments.
- Optimized query performance and data storage costs through partitioning, compression, and caching strategies in Redshift and Snowflake.
- Integrated machine learning workflows for churn prediction and customer lifetime value using MLflow and AWS SageMaker pipelines.
- Deployed observability and monitoring frameworks with Prometheus and Grafana, enabling proactive detection of data and infrastructure anomalies.
- Provided architectural leadership across distributed teams, aligning data engineering best practices with business and product scalability goals.

Lead Data Engineer (E-Commerce / Retail & Consumer Data Platforms)

11 / 2019 – 05 / 2023

RetailMetrix

- Architected and deployed data pipelines using Spark, Kafka, and Airflow, consolidating multi channel e-commerce data for analytics and personalization.
- Led modernization of legacy ETL systems to serverless AWS architectures, improving scalability and reducing infrastructure costs by 50%.
- Spearheaded implementation of a Snowflake-based data warehouse, accelerating reporting for sales, customer segmentation, and product performance.
- Developed real time inventory tracking pipelines using AWS Kinesis and Redshift Spectrum, reducing stock discrepancies by 35%.
- Partnered with business teams to create 360° customer views leveraging DBT, Looker, and BigQuery, driving data driven marketing strategies.
- Automated data validation and anomaly detection workflows using PySpark and Great Expectations, improving trust in retail analytics outputs.
- Orchestrated CI/CD pipelines for data deployments using Terraform, Jenkins, and GitHub Actions, enhancing reliability and deployment velocity.
- Mentored cross functional teams on data architecture, DevOps practices, and cloud data governance, driving a culture of operational excellence.

Senior Data Engineer (HealthTech / Healthcare Analytics)

04 / 2017 – 10 / 2019

Cohere Health

- Designed and implemented HIPAA compliant ETL frameworks using Databricks, PySpark, and Airflow, automating ingestion of patient and clinical data from multiple systems.
- Built data lakehouse architecture on AWS S3 and Delta Lake, ensuring secure and governed access to healthcare analytics and research data.
- Developed real time data ingestion pipelines using Kafka and Kinesis, enabling proactive patient monitoring and faster insights into clinical trends.
- Collaborated with analytics teams to build predictive models for patient outcomes using Python, Scikit learn,

and TensorFlow, improving care efficiency.

- Integrated FHIR and HL7 data standards into ETL processes, enhancing interoperability across EHR systems and external data sources.
- Implemented data quality and validation layers using Great Expectations, achieving 99% accuracy in clinical and operational data pipelines.
- Partnered with data governance teams to establish audit trails and encryption standards across AWS, maintaining compliance with HIPAA and GDPR.
- Optimized performance of Snowflake based analytics workloads through clustering, caching, and optimized schema design for high volume health data queries.

Data Engineer (FinTech / Digital Financial Services)

01 / 2015 – 02 / 2017

CoastPay

- Built scalable financial data infrastructure supporting secure, high volume digital payments.
- Automated ETL workflows with PySpark, Airflow, and AWS Glue, improving throughput by 40%.
- Integrated real time payment data into Redshift and Snowflake for unified analytics.
- Managed S3 data lake with automated metadata using Glue Catalog and DBT.
- Implemented data lineage and validation frameworks, reducing reporting errors by 30%.
- Developed fraud detection pipelines with Kafka, Spark Streaming, and AWS Lambda.
- Deployed credit risk and anomaly detection models using SageMaker.
- Optimized SQL and Redshift queries, improving dashboard performance by 45%.

KEY PROJECTS

Cloud Data Lakehouse Modernization

Tools: AWS S3, Snowflake, Databricks, Apache Iceberg, Airflow, DBT

- Architected a unified cloud lakehouse platform integrating multi tenant SaaS data. Enhanced query performance by 70% and reduced storage costs by 40% through optimized partitioning and compression.

E-Commerce Personalization & Analytics Platform

Tools: Apache Spark, Kafka, AWS Kinesis, Redshift, DBT, Looker

- Designed real time streaming pipelines aggregating customer, sales, and product data. Enabled personalized recommendations and improved marketing ROI by 25% through data-driven targeting models.

Healthcare Predictive Analytics & ML Pipeline

Tools: Databricks, PySpark, TensorFlow, Scikit-learn, Delta Lake, Airflow

- Built HIPAA compliant ML workflows predicting patient outcomes and readmission risks. Improved model accuracy by 15% and accelerated clinical insights delivery through automated MLOps pipelines.

Financial Transaction Intelligence System

Tools: AWS Glue, PySpark, Redshift, Kafka, SageMaker

- Developed fraud detection and credit risk scoring pipelines with real time analytics. Reduced fraud loss incidents by 35% while ensuring regulatory data lineage and reporting compliance.

EDUCATION

Bachelor of Science in Computer Science